

applications that current computer users have become familiar with.

It is not my intent to turn you all into game programmers—even though that might be a fun outcome. Rather, I'll use the pygame environment to make certain object-oriented programming techniques clearer and more visual. By working with pygame to make objects visible in a window and dealing with a user interacting with those objects, you should gain a deeper understanding of how to effectively use OOP techniques.

This chapter provides a general introduction to pygame, so most of the information and examples in this chapter will use procedural coding. Starting with the next chapter, I will explain how to use OOP effectively with pygame.

Installing Pygame

Pygame is a free downloadable package. We'll use the package manager *pip* (short for *pip installs packages*) to install Python packages. As mentioned in the Introduction, I am assuming that you have installed the official version of Python from *python.org*. The pip program is included as part of that download, so you should already have it installed.

Unlike a standard application, you must run pip from the command line. On a Mac, start the Terminal application (located in the *Utilities* subfolder inside the *Applications* folder). On a Windows system, click the Windows icon, type `cmd`, and press ENTER.

NOTE

This book was not tested with Linux systems. However, most, if not all, of the content should work with minimal tweaking. To install pygame on a Linux distribution, open a terminal in whatever way you're used to.

Enter the following commands at the command line:

```
python3 -m pip install -U pip --user
python3 -m pip install -U pygame --user
```
